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Lumbar Fusion

ORG: S-820 (ISC)
Link to Codes

MCG Health
Inpatient & Surgical Care
28th Edition

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Care Planning - Inpatient Admission and Alternatives

Clinical Indications for Procedure

- Procedure is indicated for **1 or more** of the following(1)(2)(3):
  - Spinal fracture repair, as indicated by **1 or more** of the following(6)(13):
    - Spinal instability (eg, burst fracture)
    - Neural compression
  - Spinal stenosis surgery requiring stabilization with fusion, as indicated by **ALL** of the following(7)(14)(15):
    - Unacceptable postoperative instability is judged to be likely due to extent of disease or surgery (eg, multiple vertebral levels involved).
    - Lumbar spinal stenosis treatment needed, as indicated by **1 or more** of the following:
      - Rapidly progressive or very severe symptoms of neurogenic claudication with imaging findings of lumbar spinal stenosis that correlate to clinical findings(16)
      - Leg or buttock neurogenic claudication symptoms and **ALL** of the following:
        - Symptoms that are persistent and disabling
        - Imaging findings of lumbar spinal stenosis that correlate with clinical findings
        - Failure of 3 months of nonoperative therapy
  - Lumbar spondylolisthesis, as indicated by **1 or more** of the following(8)(17)(18)(19):
    - Rapidly progressive or very severe neurologic deficits (eg, bowel or bladder dysfunction)
    - Symptoms requiring treatment, as indicated by **ALL** of the following:
      - Patient has persistent disabling symptoms, including **1 or more** of the following:
        - Low back pain
        - Neurogenic claudication
        - Radicular pain
      - Treatment is indicated by **ALL** of the following:
        - Listhesis demonstrated on imaging

        - Symptoms correlate with findings on MRI or other imaging.
        - Failure of 3 months of nonoperative therapy
      - Child or adolescent with high-grade (greater than 50% slippage) spondylolisthesis (to prevent progression)(20)(21)
    - Adjacent segment disease, as indicated by **ALL** of the following(9)(22):
      - Radiographic evidence of adjacent segment disease (eg, neural compression) that correlates with symptoms
      - Persistent disabling symptoms (low back pain, radiculopathy)
      - Failure of 3 months of nonoperative therapy
    - Lumbar pseudarthrosis and **ALL** of the following(10):
      - Symptoms (eg, pain) unresponsive to nonoperative therapy
      - Lumbar imaging findings consistent with symptoms
      - Alternative etiologies of symptoms ruled out
  - Spinal repair with fusion (eg, for instability due to extensive surgery) in conjunction with other procedures (eg, laminectomy) for neural decompression, fracture, dislocation, infection, abscess, tumor, or coronal imbalance(11)(12)(23)(24)

Alternatives to Procedure

- Alternatives include(1)(8)(13)(15)(25):
  - Nonoperative measures, including(17)(20):
    - Anti-inflammatory medication
    - Analgesics
    - Lumbar bracing
    - Physical therapy
    - Activity modification
  - Laminectomy without fusion. See Lumbar Laminectomy ISC guideline.
  - Diskectomy without fusion. See Lumbar Diskectomy, Foraminotomy, or Laminotomy ISC guideline.
  - Hemilaminectomy
  - Multilevel unilateral or bilateral laminotomy
  - Fracture fixation without fusion(6)
  - Radiation therapy, chemotherapy, and glucocorticoids (eg, neoplastic epidural compression)(24)

Operative Status Criteria

Goal Length of Stay: 2 days postoperative

- Inpatient

Preoperative Care Planning

- Preoperative care planning needs may include(1)(8):
  - Routine preoperative evaluation. See Preoperative Education, Assessment, and Planning Tool SR.
  - Diagnostic test scheduling, including(28)(29)(30):
    - Radiographs
    - Imaging (eg, MRI or CT myelogram)
    - Electromyography
    - Bone mineral density(31)(32)
  - Preoperative treatment, procedures, and stabilization, including:
    - Smoking cessation counseling(33)(34)
    - Selection of bone graft site or material(31)(35)
    - Physical and occupational therapy consultation for development of rehabilitation plan, including progressive exercises, muscle strengthening, and activity pacing
    - Fitting for cast or brace
    - Cessation or reduction of opioid use
  - Preoperative discharge planning as appropriate. See Discharge Planning in this guideline.

Hospitalization

Optimal Recovery Course

Day	Level of Care	Clinical Status	Activity	Routes	Interventions	Medications

1	• OR to floor • Social Determinants of Health Assessment • Readmission Risk Assessment • Discharge planning	• **Clinical Indications met[A]**	• Possibly assisted, limited ambulation postoperatively	• IV fluids • IV medications	• DVT prophylaxis[B]	• Prophylactic antibiotics[C] • Multimodal analgesia, possible PCA
2	• Floor • Social Determinants of Health Assessment • Readmission Risk Assessment	• **Hemodynamic stability** • **No new neurologic deficit**	• **Progressive ambulation**	• **Oral hydration** • Parenteral or oral medications • Diet as tolerated	• **NG tube absent** • DVT prophylaxis	• Multimodal analgesia
3	• Social Determinants of Health Assessment • Readmission Risk Assessment • Floor to discharge • Complete discharge planning	• **Hemodynamic stability** • **No new neurologic deficit** • **No evidence of postoperative or surgical site infection** • **Voiding ability at baseline** • **Adequate ambulation** • **Pain absent or managed** • **Discharge plans and education understood**	• **Ambulatory or acceptable for next level of care[D]**	• **Oral hydration[E]** • **Oral medications or regimen acceptable for next level of care** • **Oral diet or acceptable for next level of care**	• DVT prophylaxis	• **PCA absent[F]**
(1)(8)(13)(38)N						

Recovery Milestones are indicated in **bold**.

Goal Length of Stay: 2 days postoperative

Note: Goal Length of Stay assumes optimal recovery, decision making, and care. Patients may be discharged to a lower level of care (either later than or sooner than the goal) when it is appropriate for their clinical status and care needs.

Extended Stay

Minimal (a few hours to 1 day), Brief (1 to 3 days), Moderate (4 to 7 days), and Prolonged (more than 7 days).

- Extended stay beyond goal length of stay may be needed for(38):
  - Failure to meet discharge criteria (recovery milestones within final day in Optimal Recovery Course)
    - Expect brief stay extension.
  - Complications of surgery (eg, prolonged ileus, respiratory insufficiency)(39)
    - Expect brief stay extension.
  - Pathologic fracture or neoplasm
    - Expect brief stay extension.
  - Spinal abscess or osteomyelitis
    - Expect brief to moderate stay extension.
  - Severe preoperative deficit or injury

        - Patient with significant neurologic compromise or multiple injuries will require longer acute care and recovery times.
        - Stay extension varies depending on injury.
    - Dural tear(40)
        - Anticipate possible CSF drainage and surgical repair.
        - Expect brief to moderate stay extension.
    - Intraspinal hemorrhage
        - Intraspinal hemorrhage may need surgical repair.
        - Expect brief to moderate stay extension.
    - Extraspinal hemorrhage
        - Anticipate possible evacuation and surgical repair.
        - Expect brief to moderate stay extension.
    - Nerve root or cauda equina injury
        - Anticipate possible physical therapy and rehabilitation.
        - Expect brief to moderate stay extension.
    - Visual loss (eg, ischemic optic retinopathy)
        - Expect brief to moderate stay extension.
    - Combined (anterior and posterior) procedures
        - Expect brief stay extension.
    - Active comorbid illness (eg, cardiovascular, renal, or pulmonary disease)(41)
        - Anticipate relevant monitoring, treatment, and consultation.
        - Expect brief stay extension.
    - Postoperative altered mental status (eg, delirium)
        - Anticipate evaluation for etiology (eg, electrolyte abnormalities, pain, medications) and specific management as appropriate.
        - Expect brief stay extension.

See Common Complications and Conditions ISC for further information.

Hospital Care Planning

- Hospital evaluation and care needs may include(1)(8)(13):
    - Treatment and procedure scheduling and completion, including:
        - IV antibiotics
        - DVT prophylaxis(36)
        - Tranexamic acid(44)
        - Transfusion
        - Multimodal analgesia (45)(46)
    - Consultation, assessment, and other services scheduling and completion, including:
        - Physical therapy(47)
        - Occupational therapy
        - Bracing or casting
    - Identification of patient at high risk for readmission to prioritize transition and post-acute care(42):
        - Risk of readmission is increased by presence of **1 or more** of the following(48)(49)(50)(51)(52)(53)(54)(55)(56):
            - Hospitalization (nonelective) in past 6 months(57)(58)(59)
            - No source of outpatient care other than emergency department (eg, no primary care provider)(59)(60)
            - Severe care transition barriers (eg, no caregiver, homeless)(57)(58)(61)
            - Severe or end-stage renal disease (on dialysis or GFR less than 30 mL/min/1.73m^2 (0.5 mL/sec/1.73m^2))(57)(62) eGFR - Adult Calculator eGFR - Pediatric Calculator
            - Advanced liver disease (eg, cirrhosis with portal hypertension, history of variceal bleed)
            - Postoperative pneumonia(42)(43)
            - Postoperative wound infection(42)(43)(63)
    - Monitoring patient's status for deterioration and comorbid conditions (see Inpatient Monitoring and Assessment Tool SR); key items include:
        - Neurovascular status of lower extremities
        - Pain management(64)
        - Hemodynamic stability
        - New-onset headache suspicious for dural tear
        - Urinary retention
        - Wound management, observing for healing at spine or for hematoma and signs of infection at donor site
        - Drain output amount and type

Discharge

Discharge Planning

- Discharge planning includes[G]:
  - Assessment of needs and planning for care, including(66)(67):
    - Develop and modify treatment plan (involving multiple providers) as needed.
    - Evaluate and address preadmission functioning as needed.
    - Evaluate and address psychosocial status issues as indicated. See Psychosocial Assessment SR for further information.
    - Evaluate and address social determinants of health (eg, housing, food). See Social Determinants of Health Screening Tool SR for further information.(65)(68)
    - Evaluate and address patient or caregiver preferences as indicated.
    - Identify skilled services needed at next level of care, with specific attention to:
      - Medication management, adherence instruction, and side effects assessment(69)
      - Neurologic status assessment(70)
      - Pain management(70)
      - Rehabilitation therapy or equipment coordination(71)
      - Wound or dressing management(13)
  - Early identification of anticipated discharge destination; options include(67)(72):
    - Home, considerations include:
      - Home safety assessment. See Home Safety Assessment SR for further information.
      - Patient safe to go home; examples include(73)(74)(75):
        - Medical status stable for patient's condition
        - Functional care can safely be provided with available resources.
        - Mental status stable for patient's condition
        - Medication availability confirmed and reconciliation complete
        - Patient/caregiver education completed with written discharge instructions provided
        - Community resources identified and referrals made, as needed
        - Home care arranged, if indicated
        - Necessary medical equipment delivery arranged or available in home, if indicated
        - Necessary medical supplies ordered, or patient/caregiver can obtain, if indicated
      - Access to follow-up care
      - Self-management ability if appropriate. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment SR for further information.
      - Caregiver need, ability, and availability
    - Post-acute skilled care or custodial care as indicated. See Discharge Planning Tool SR for further information.
  - Transitions of care plan complete, including(67)(72)(76):
    - Patient and caregiver education complete. See Lumbar Fusion: Patient Education for Clinicians SR for further information.
      - See Teach Back Tool SR for further information.
    - Medication reconciliation completion includes(69)(77):
      - Compare patient's discharge list of medications (prescribed and over-the-counter) against provider's admission or transfer orders.
      - Assess each medication for correlation to disease state or medical condition.
      - Report medication discrepancies to prescribing provider, attending physician, and primary care provider, and ensure accurate medication order is identified.
      - Provide reconciled medication list to all treating providers.
      - Confirm that patient or caregiver can acquire medication.
      - Educate patient and caregiver.
        - Provide complete medication list to patient and caregiver.
        - Importance of presenting personal medication list to all providers at each care transition, including all provider appointments
        - Reason, dosage, and timing of medication (eg, use "teach-back" techniques)(78)
      - Encourage communication between patient, caregiver, and pharmacy for obtaining prescriptions, setting up home medication delivery, and reviewing for drug-drug interactions.
      - See Medication Reconciliation Tool SR for further information.
    - Plan communicated to patient, caregiver, and all members of care team, including(79):
      - Inpatient care and service providers
      - Primary care provider
      - All post-discharge care and service providers
    - Appointments planned or scheduled, which may include:

        - Primary care provider
        - Neurosurgeon(80)
        - Orthopedic surgeon
        - Rehabilitation therapy services
        - Specialists for management of comorbidities as needed
        - Other
    - Outpatient testing and procedure plans made, which may include:
        - Laboratory testing
        - Other
    - Referrals made for assistance or support, which may include:
        - Financial, for follow-up care, medication, and transportation
        - Tobacco use treatment(81)
        - Vocational rehabilitation(82)
        - Other
    - Medical equipment and supplies coordinated (ie, delivered or delivery confirmed), which may include:
        - Ambulation devices (eg, cane, crutches, walker)
        - Antiembolic or compression stockings
        - Immobilizers (eg, braces, splints)
        - Wound care equipment and supplies(83)
        - Other

Discharge Destination

- Post-hospital levels of admission may include:
    - Home.
    - Home healthcare. See Home Care Indications for Admission Section HC in Lumbar Spine Surgery guideline in Home Care.
    - Recovery facility care. See Recovery Facility Care Indications for Admission Section RFC in Lumbar Spine Surgery guideline in Recovery Facility Care.

Evidence Summary

Background

Lumbar fusion may be performed either directly through an open incision or endoscopically based on patient characteristics and surgeon experience.(4)(5) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 19 published peer reviewed articles and 5 book sections.

For spinal fracture repair, a narrative review states that lumbosacral fusion is indicated for patients with unstable sacral fractures.(2) **(EG 2)** An orthopedic surgery textbook states that lumbar fusion surgery is appropriate for patients with spinal fractures associated with neural compression or instability.(6) **(EG 2)**

For spinal stenosis surgery, an orthopedic surgery textbook notes that additional lumbar fusion may be needed if excessive bony resection occurs which destabilizes the spine.(1) **(EG 2)** A narrative review states that interbody fusion of the lumbar spine can be used in the setting of spinal stenosis to provide indirect decompression and improve symptoms.(7) **(EG 2)**

For lumbar spondylolisthesis, an orthopedic surgery textbook states that lumbar fusion is an appropriate treatment for spondylolisthesis of all types when patients have chronic pain from instability.(1)(8) **(EG 2)** Narrative reviews note that lumbar fusion is appropriate for patients with high-grade spondylolisthesis.(2)(3)(9) **(EG 2)**

For adjacent segment disease, a narrative review notes that lateral lumbar fusion can be a helpful procedure to improve symptoms.(9) **(EG 2)**

For lumbar pseudarthrosis, an orthopedic surgery textbook and a narrative review state that repeat lumbar fusion is a recommended procedure to improve symptoms in patients with localized pain over the fusion area and when lumbar imaging findings are consistent with the symptoms.(1)(10) **(EG 2)**

For spinal repair with fusion in conjunction with other procedures, an orthopedic surgery textbook states that fusion can be a helpful adjunctive procedure when any significant amount of bone is resected due to tumor treatment.(11) **(EG 2)** A narrative review states that surgical management of spinal tuberculosis includes debridement, decompression, and adequate fusion to provide for early mobilization.(12) **(EG 2)**

Alternatives

A randomized controlled study of 233 patients (mean age 67 years) undergoing surgery for lumbar spinal stenosis (at 1 or 2 levels) found that the addition of fusion to laminectomy did not result in improved clinical outcomes at 2-year and 5-year follow-up, and that fusion resulted in longer operative times and more blood loss.(26) **(EG 1)** In the same trial, when patients were stratified by the presence or absence of spondylolisthesis, the results were similar.(26) **(EG 1)** A study of 211 patients (mean age 66 years) with lumbar spinal stenosis randomized to 1-level or 2-level decompression alone or decompression with fusion found that, although spinal restenosis was less frequent after decompression with fusion at the operative level (4% vs 14% for decompression with fusion vs decompression alone, respectively), this was more than offset by an increase in the rate of new stenosis at an adjacent level (44% vs 17%).(27) **(EG 1)** A randomized noninferiority trial compared lumbar decompression alone with lumbar decompression with instrumental fusion in 262 patients (mean age 66 years) with leg or back pain due to lumbar spinal stenosis and single-level spondylolisthesis of 3 mm or more that has persisted despite nonoperative treatment (most patients with symptoms for more than 1 year), and concluded there was noninferiority of decompression alone in patient-reported outcomes (eg, Oswestry Disability Index) at 2-year follow-up.(18) **(EG 1)** In the same trial, the decompression alone group had a slightly higher reoperation rate (12.5% vs 9.1%).(18) **(EG 1)**

Hospitalization

Readmission risk and reduction: Multivariate analysis of 2042 patients undergoing single-level lumbar interbody fusion found that postoperative complications of pneumonia or wound infection were independently associated with an increased risk of readmission within 30 days.(42) **(EG 2)** Analysis of a national database of patients who underwent lumbar interbody fusion (23,336 patients) found, after multivariate analysis, that the presence of any surgical complication (eg, wound dehiscence, infection) was independently associated with an increased risk of 30-day readmission.(43) **(EG 2)**

Length of Stay

Analysis of national hospital discharge data for a pediatric population shows 48% of hospitalized patients undergoing a principal procedure of lumbar fusion discharged in 2 days or less.(38) **(EG 3)** Analysis of national hospital discharge data shows 44% of hospitalized adult patients undergoing a principal procedure of lumbar fusion discharged in 2 days or less.(38) **(EG 3)**

Rationale

Use of this MCG care guideline helps the clinician identify, for a given procedure, which patient-specific factors and clinical conditions are appropriate for that procedure. The evidence-based clinical indication criteria assist the clinician in the decision to appropriately perform a procedure, evaluating whether the potential benefits of a procedure outweigh the potential risks. For Medicare enrollees, surgical MCG care guidelines also identify which procedures CMS has designated as inpatient only.

Use of these evidence-based clinical criteria to support decision making around the need for a given procedure is of benefit to the patient, as all procedures come with inherent risk that must be balanced by anticipated clinical benefit. Utilizing evidence-based clinical criteria enables a more accurate and patient-specific decision-making process. In addition, the use of evidence-based guidelines can help reduce unwarranted variation in care, such as divergent clinical thresholds to perform a procedure for clinically similar patients that vary across geographic regions, between facilities, and among individual clinicians.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 412.3(84); 42 CFR 419.22(85); 42 CFR 422.101(86)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 1 - Inpatient Hospital Services Covered Under Part A(87); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 6 - Hospital Services Covered Under Part B(88); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(89); CMS IOM Publication 100-08, Medicare Program Integrity Manual, Chapter 6, Section 6.5 - Medical Review of Inpatient Hospital Claims for Part A Payment(90)

Medicare Coverage Determinations: Medicare Coverage Database(91)

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Footnotes

[A] See Clinical Indications for Procedure in this guideline. [A in Context Link 1]

[B] Low-dose unfractionated heparin, low-molecular-weight heparin, or intermittent pneumatic compression devices may be appropriate.(36) [B in Context Link 1]

[C] Prophylactic antibiotics may be considered to decrease the rate of infections following instrumented spine fusion.(37) [C in Context Link 1]

[D] Patient is ambulatory or near baseline activity for age and development. [D in Context Link 1]

[E] Some patients may have their hydration needs met via alternative means (eg, percutaneous endoscopic gastrostomy tube). [E in Context Link 1]

[F] Use Multimodal analgesia or individual analgesic agent as indicated. [F in Context Link 1]

[G] Discharge instructions should be given in the patient's and caregiver's native language using trained language interpreters whenever possible.(65) [G in Context Link 1]

Definitions

Hemodynamic stability

- Hemodynamic stability, as indicated by **1 or more** of the following:
  - Hemodynamic abnormalities at baseline or acceptable for next level of care
  - Patient hemodynamically stable, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
    - Tachycardia absent
    - Hypotension absent
    - No evidence of inadequate perfusion (eg, no myocardial ischemia)
    - No other hemodynamic abnormalities (eg, no Orthostatic hypotension)

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Hypotension absent

- Hypotension absent[A], as indicated by **1 or more** of the following(1)(2)(3)(4):
  - SBP greater than or equal to 90 mm Hg in adult or child 10 years or older
  - Mean arterial pressure[B] greater than or equal to 70 mm Hg in adult or child 10 years or older

- Mean arterial pressure[B] at patient's baseline (eg, healthy adult with low SBP), at intentional therapeutic goal (eg, patient with heart failure), or acceptable for next level of care (eg, blood pressure stable and no significant signs or symptoms due to low blood pressure)
- SBP greater than or equal to sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age
- SBP greater than or equal to 70 mm Hg in infant 1 to 11 months of age

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Footnotes

A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = 1/3 SBP + 2/3 DBP.

Multimodal analgesia

- Multimodal analgesia involves the utilization of 2 or more analgesic agents with different mechanisms of action in order to provide additive or synergistic pain control, while minimizing side effects and reliance on opioids.(1)(2)(3)

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Orthostatic hypotension

- Orthostatic hypotension,[A][B] as indicated by **1 or more** of the following(1)(2)(3):
  - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
  - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

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Footnotes

A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension which relies upon blood pressure alone.(1)(2)(3)
B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one

time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Readmission Risk Assessment

- Risk of readmission is increased by presence of **1 or more** of the following(1)(2)(3)(4)(5)(6)(7)(8)(9):
  - Hospitalization (nonelective) in past 6 months(10)(11)(12)
  - No source of outpatient care other than emergency department (eg, no primary care provider)(12)(13)
  - Severe care transition barriers (eg, no caregiver, homeless)(10)(11)(14)
  - Severe or end-stage renal disease (on dialysis or GFR less than 30 mL/min/1.73m^2 (0.5 mL/sec/1.73m^2))(10)(15)
    eGFR - Adult Calculator eGFR - Pediatric Calculator
  - Advanced liver disease (eg, cirrhosis with portal hypertension, history of variceal bleed)
  - Postoperative pneumonia(16)(17)
  - Postoperative wound infection(16)(17)(18)

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Social Determinants of Health Assessment

- Risk of poor health outcomes may be increased by the presence of **1 or more** of the following social determinants of health(1)(2)(3)(4):

- Housing insecurity, as indicated by **1 or more** of the following:
  - Individual or caregiver's current living situation is **1 or more** of the following(5):
    - Does not have own housing (eg, staying in a hotel, shelter, or with others)
    - Has own housing (eg, house, apartment), but at risk of losing it in the future (ie, behind on rent or mortgage)
    - Has own housing (eg, house, apartment), but has lived in 3 or more places in past year
  - Current housing has **1 or more** of the following:
    - Electrical appliances (eg, stove, refrigerator) not working or unavailable
    - Insufficient heating or cooling
    - Insufficient ventilation
    - Lead paint or pipes
    - Mold
    - Pests (eg, bugs) or rodents
    - Smoke detectors not working or unavailable
- Food insecurity, as indicated by **1 or more** of the following(6):
  - In the past year, individual or caregiver ran out of food and did not have money to buy more food.
  - In the past year, individual or caregiver worried that they would run out of food before they received money to buy more food.
- Insufficient transportation, as indicated by **1 or more** of the following(7):
  - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of transportation.
  - In the past year, individual or caregiver missed nonmedical activities, work, or could not get things needed for daily living due to lack of transportation.
- Insufficient utilities, as indicated by **1 or more** of the following(8):
  - Utilities (eg, electricity, water, gas, or oil) are currently shut off or unavailable.
  - In the past year, electric, water, gas, or oil company threatened to shut off services.
- Personal safety risk, as indicated by **2 or more** of the following(6):
  - Individual is sometimes or frequently physically hurt by another person (including family member).
  - Individual is sometimes or frequently insulted or talked down to by another person (including family member).
  - Individual is sometimes or frequently threatened with physical harm by another person (including family member).
  - Individual is sometimes or frequently screamed or cursed at by another person (including family member).
- Insufficient dependent care, as indicated by **1 or more** of the following:
  - In the past year, individual or caregiver was unable to work due to lack of dependent care.
  - In the past year, individual or caregiver was unable to work more (additional) hours due to lack of dependent care.
  - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of dependent care.
  - In the past year, individual or caregiver missed nonmedical activities (eg, school, church, social activity) due to lack of dependent care.
- Depression risk, as indicated by **ALL** of the following:
  - In the past 2 weeks, individual had little interest or pleasure in normal activities on at least several days.
  - In the past 2 weeks, individual felt down, depressed, or hopeless on at least several days.

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Tachycardia absent

- Tachycardia[A][B] absent, as indicated by **1 or more** of the following(1)(2):
  - Heart rate less than or equal to 100 beats per minute in adult or child 6 years or older
  - Heart rate less than or equal to 115 beats per minute in child 3 to 5 years of age
  - Heart rate less than or equal to 125 beats per minute in child 1 or 2 years of age
  - Heart rate less than or equal to 130 beats per minute in infant 6 to 11 months of age
  - Heart rate less than or equal to 150 beats per minute in infant 3 to 5 months of age
  - Heart rate less than or equal to 160 beats per minute in infant 1 or 2 months of age

References

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Footnotes

A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Codes

ICD-10 Diagnosis: A18.01, C41.2, C79.40, C79.49, D16.6, D32.1, D33.4, D42.1, D43.4, D48.0, D49.2, G06.1, M43.05, M43.06, M43.07, M43.09, M43.15, M43.16, M43.17, M43.19, M46.45, M46.46, M46.47, M47.15, M47.16, M47.25, M47.26, M47.27, M47.815, M47.816, M47.817, M47.895, M47.896, M47.897, M48.05, M48.061, M48.062, M48.07, M51.05, M51.06, M51.15, M51.16, M51.17, M51.25, M51.26, M51.27, M51.35, M51.36, M51.37, M51.85, M51.86, M51.87, M51.A0, M51.A1, M51.A2, M51.A3, M51.A4, M51.A5, M54.15, M54.16, M54.17, M96.1, M99.13, M99.23, M99.33, M99.43, M99.53, M99.63, M99.73, Q76.2, S33.0XXA, S33.100A, S33.101A, S33.110A, S33.111A, S33.120A, S33.121A, S33.130A, S33.131A, S33.140A, S33.141A, S34.101A, S34.102A, S34.103A, S34.104A, S34.105A, S34.109A, S34.111A, S34.112A, S34.113A, S34.114A, S34.115A, S34.119A, S34.121A, S34.122A, S34.123A, S34.124A, S34.125A, S34.129A [Hide]

ICD-10 Procedure: 0RGA070, 0RGA071, 0RGA07J, 0RGA0A0, 0RGA0AJ, 0RGA0J0, 0RGA0J1, 0RGA0JJ, 0RGA0K0, 0RGA0K1, 0RGA0KJ, 0RGA370, 0RGA371, 0RGA37J, 0RGA3A0, 0RGA3AJ, 0RGA3J0, 0RGA3J1, 0RGA3JJ, 0RGA3K0, 0RGA3K1, 0RGA3KJ, 0SG0070, 0SG0071, 0SG007J, 0SG00A0, 0SG00AJ, 0SG00J0, 0SG00J1, 0SG00JJ, 0SG00K0, 0SG00K1, 0SG00KJ, 0SG0370, 0SG0371, 0SG037J, 0SG03A0, 0SG03AJ, 0SG03J0, 0SG03J1, 0SG03JJ, 0SG03K0, 0SG03K1, 0SG03KJ, 0SG1070, 0SG1071, 0SG107J, 0SG10A0, 0SG10AJ, 0SG10J0, 0SG10J1, 0SG10JJ, 0SG10K0, 0SG10K1, 0SG10KJ, 0SG1370, 0SG1371, 0SG137J, 0SG13A0, 0SG13AJ, 0SG13J0, 0SG13J1, 0SG13JJ, 0SG13K0, 0SG13K1, 0SG13KJ, 0SG3070, 0SG3071, 0SG307J, 0SG30A0, 0SG30AJ, 0SG30J0, 0SG30J1, 0SG30JJ, 0SG30K0, 0SG30K1, 0SG30KJ, 0SG3370, 0SG3371, 0SG337J, 0SG33A0, 0SG33AJ, 0SG33J0, 0SG33J1, 0SG33JJ, 0SG33K0, 0SG33K1, 0SG33KJ, XRGA0R7, XRGA3R7, XRGB0R7, XRGB3R7, XRGC0R7, XRGC3R7, XRGD0R7, XRGD3R7 [Hide]

CPT®: 22533, 22534, 22558, 22585, 22586, 22612, 22630, 22632, 22633, 22634

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